

Jinquan Pan

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EDUCATION

Yale University, New Haven, CT Expected May 2026

M.S. in Computer Science, *Cumulative GPA*: 4.0/4.0

Relevant Coursework: Theory of Distributed System, Blockchain and Decentralized system, Software Analysis

Leadership: Head Course Manager, Computer Science classes; Member, Yale Engineering Club

Boston University, Boston, MA Jan 2024

B.A. in Mathematics and Computer Science, *Cumulative GPA*: 3.92/4.0 (Magna Cum Laude)

Relevant Coursework: Software Development, Data Mechanics, Advanced Machine Learning, Operating Systems

Leadership: Associate Principal, Technical Department, BUCSSA; Senior Team Lead, Hack4Impact

TECHNICAL SKILLS

- *Computer Languages*: Python, Java, C, Go, PHP, SQL, HQL, HTML, CSS
- *Tools*: AWS (Lambda, DynamoDB, S3, EC2, ECS, CloudFormation), Kubernetes, Docker, Terraform, Apache Spark, Kafka, gRPC, Redis, Django, Flask, FastAPI, MySQL, PostgreSQL
- *Areas of Expertise*: Distributed Systems, API Design & Development, Scalable Microservices, Cloud Computing, CI/CD Pipelines, High-Throughput Data Processing, System Performance Optimization, Unit & Integration Testing

RESEARCH & ENGINEER EXPERIENCE

Big Data Engineer, **Transwarp Technology (Shanghai) Co. Ltd**, Guangdong, China Mar 2024 – Jul 2024

- Architected a distributed, high-availability data pipeline using Spark and Kafka, processing terabytes of financial data daily with 99.99% uptime, reducing compute latency by 60% while ensuring auto-recovery and horizontal scaling.
- Developed high-performance data APIs for financial hedging efficiency and Granger causality metrics, accelerating response time by 4x and enabling real-time risk analysis for \$10M+ daily trades.
- Led full-cycle development of a scalable, high-availability trading database, managing interns and collaborating with stakeholders. Replaced legacy infrastructure, enabling the department to adopt the new system for daily operations.

Cloud Security Engineer, **Red Hat & Microsoft**, Cambridge, MA Sep 2023 – Jan 2024

- Built a secure, multi-tenant Spark & YARN cluster on Kubernetes with Kata Containers, improving workload isolation and reducing processing failures by 30%, while ensuring 99.9% system uptime.
- Optimized distributed Spark workloads, leveraging iperf3 and fio benchmarking tools to fine-tune network and disk performance, cutting batch job execution time by 10x and improving fault recovery speed by 40%.

Teaching Assistant, **Boston University Department of Computer Science**, Boston, MA Sep 2022 – Jan 2024

- Led security lab sessions for 200+ students, designing hands-on exercises in encryption, authentication, and secure web protocols. Mentored top students, helping them develop advanced cybersecurity projects.
- Developed and deployed two secure challenge websites using PHP, HTML, Python, JavaScript, and MySQL, simulating real-world web security vulnerabilities. Created SQL injection, XSS, and authentication bypass scenarios to improve students' cybersecurity skills.

SELECTED PROJECTS

AI-Powered Regulatory Comment Generator, New Haven, CT Oct 2024 – Current

- Designed and built a scalable, event-driven AI platform using Python, Flask, PostgreSQL, and Kafka, processing 2M+ of Federal Register entries for real-time regulatory tracking and automated public comment generation.
- Developed a multi-tenant API service with customizable branding, multilingual support, and plugin-based extensibility, enabling seamless integration for advocacy organizations and automating legal document generation.

CS Education Equity Dashboard & ML Insights, Boston, MA May 2023 – July 2023

- Built a real-time analytics dashboard using Flask, React, and Redis, enabling dynamic data visualization and exploration of disparities in computer science education across 300+ Massachusetts school districts.
- Developed and deployed machine learning models with scikit-learn and Pandas, analyzing key demographic and policy factors affecting education access and generating data-driven insights for policymakers.